

- 3 (a) (i) (tangent to line gives) direction of force on a (small test) mass B1 [1]
- (ii) (tangent to line gives) direction of force on a (small test) charge M1
charge is positive A1 [2]
- (b) similarity:
e.g. radial fields
lines normal to surface
greater separation of lines with increased distance from sphere
field strength $\propto 1 / (\text{distance to centre of sphere})^2$
(allow any sensible answer) B1
- difference:
e.g. gravitational force (always) towards sphere B1
electric force direction depends on sign of charge on sphere / towards or
away from sphere B1
e.g. gravitational field/force is attractive (B1)
electric field/force is attractive or repulsive (B1)
(allow any sensible comparison) [3]
- (c) gravitational force = $1.67 \times 10^{-27} \times 9.81$
= $1.6 \times 10^{-26} \text{ N}$ A1
electric force = $1.6 \times 10^{-19} \times 270 / (1.8 \times 10^{-2})$ C1
= $2.4 \times 10^{-15} \text{ N}$ A1
electric force very much greater than gravitational force B1 [4]

- 4 (a) force on proton is normal to velocity and field

M1